t3-w06 — condition C2

Construction — Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

> C1 plus the lesson note as source evidence.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	**present**
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	**present**

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

- construct parallel lines
- construct perpendicular lines
- bisect a given line segment
- construct angle 90°
- construct angle 60°

entry_behaviour

- Students have been taught how to draw lines.

instructional_materials

- Mathematical set.

Objective 1 — construct parallel lines

objective_text

construct parallel lines

current_step

1

instructional_events

• present_content
• elicit_performance

student_tasks

• construct parallel lines

source_figures


```
[
 {
  "caption": "Line $AB$, with a point $P$ marked above it. A set square is drawn below $AB$ with one edge along the line: the right angle is at the left end of that edge, one leg runs down from it, and the hypotenuse closes back to a point on $AB$ to the right.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "The same set square on $AB$, with the ruler now drawn as a long strip lying along the set square's downward edge. The set square's vertical edge and the ruler's right edge are **layered** — only one line is visible on the page. The right angle marked at the corner is the set square's own, between the edge lying along $AB$ and the edge rising from it.",
  "figure_ref": "fig-02"
 },
 {
  "caption": "The same arrangement drawn a second time alongside the first — the set square's edge again **layered** on the ruler's right edge, with no gap between them — and with the ruler leaning — running from upper left to lower right across $AB$ instead of standing vertical. Only $A$ is labelled. The slanted ruler is the point of this panel — it shows that the ruler's inclination does not affect the construction.",
  "figure_ref": "fig-03"
 },
 {
  "caption": "The vertical ruler held fixed, with the set square drawn twice — once with its edge on $AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow between the two positions pointing up along the ruler.",
  "figure_ref": "fig-04"
 },
 {
  "caption": "The same slide drawn a second time alongside it, with the leaning ruler. Both set square positions are **layered** on the ruler's right edge, as in fig-03. Here the upper line carries the point $P$, and the set square's upper position has its edge running through $P$.",
  "figure_ref": "fig-05"
 }
]
```


lesson_content

```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Examples:** construct a line through Point P so that it is parallel to AB.\n\n(a)
Place a set square so that one edge is accuratly along AB\n\n**fig-01.** Line $AB$, with a point
$P$ marked above it. A set square is drawn below $AB$ with\none edge along the line: the right
angle is at the left end of that edge, one leg runs down from\nit, and the hypotenuse closes back
to a point on $AB$ to the right.\n\n[drawing attached to this section as fig-01, marked A, B, P]
\n\n(b) Place a ruler along one of the other edge of the set square. Use the left hand edge if
you\nare right handed.\n\n**fig-02.** The same set square on $AB$, with the ruler now drawn as a
long strip lying along the\nset square's downward edge. The set square's vertical edge and the
ruler's right edge are\n**layered** — only one line is visible on the page. The right angle
marked at the corner is the\nset square's own, between the edge lying along $AB$ and the edge
rising from it.\n\n[drawing attached to this section as fig-02, marked A, B]\n\n**fig-03.** The
same arrangement drawn a second time alongside the first — the set square's edge\nagain
**layered** on the ruler's right edge, with no gap between them — and with the ruler leaning\n—
running from upper left to lower right across $AB$ instead of standing vertical. Only $A$
is\nlabelled. The slanted ruler is the point of this panel — it shows that the ruler's
inclination\ndoes not affect the construction.\n\n[drawing attached to this section as fig-03,
marked A]\n\n(c) Hold the ruler firmly, slide the set square along the ruler towards P. Stop when
the edge that\nwas on AB reaches P. Draw a line along this edge of the set square through P.
\n\n**fig-04.** The vertical ruler held fixed, with the set square drawn twice — once with its
edge on\n$AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow
between the two\npositions pointing up along the ruler.\n\n[drawing attached to this section as
fig-04, marked A, B]\n\n**fig-05.** The same slide drawn a second time alongside it, with the
leaning ruler. Both set\nsquare positions are **layered** on the ruler's right edge, as in
fig-03. Here the upper\nline carries the point $P$, and the set square's upper position has its
edge running through $P$.\n\n[drawing attached to this section as fig-05, marked A, P]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — construct perpendicular lines

objective_text

construct perpendicular lines

current_step

2

instructional_events

• present_content

source_figures


```
[
 {
  "caption": "Line $AB$ with a second line $XY$ crossing it, $X$ above and $Y$ below, and a
right-angle mark in the upper right quadrant at the crossing.",
  "figure_ref": "fig-06"
 },
 {
  "caption": "A plain line with a stroke at each end, labelled $A$ on the left and $B$ on the
right. The point $Q$ named in the instruction is not marked on the drawing.",
  "figure_ref": "fig-07"
 },
 {
  "caption": "Line $AB$ with the set square standing on it as a narrow right-angled triangle: a
short base along $AB$, a near-vertical edge rising from the right end of that base, and the
hypotenuse running up from the left.",
  "figure_ref": "fig-08"
 },
 {
  "caption": "Line $AB$ with $XY$ now drawn across it, $X$ above and $Y$ below, and a right-angle
mark in the upper right quadrant. The foot of $XY$ is not labelled $Q$.",
  "figure_ref": "fig-09"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "Two lines are said to be perpendicular to each other when they meet at a right
angle i.e\n$90^\\circ$\n\n**fig-06.** Line $AB$ with a second line $XY$ crossing it, $X$ above
and $Y$ below, and a\nright-angle mark in the upper right quadrant at the crossing.\n\n[drawing
attached to this section as fig-06, marked A, B, X, Y]\n\n1. Draw a line AB mark a point Q on the
line.\n\n**fig-07.** A plain line with a stroke at each end, labelled $A$ on the left and $B$ on
the right.\nThe point $Q$ named in the instruction is not marked on the drawing.\n\n[drawing
attached to this section as fig-07, marked A, B]\n\n2. Place the edge of the set square on line
AB. Let the edge of the ruler meet the hypotenuse of\n   the set square as shown.\n\n**fig-08.**
Line $AB$ with the set square standing on it as a narrow right-angled triangle: a\nshort base
along $AB$, a near-vertical edge rising from the right end of that base, and the\nhypotenuse
running up from the left.\n\n[drawing attached to this section as fig-08, marked A, B]\n\n3.
Pressing firmly on the ruler, slide the set square down along the hypotenuse side until the\n
third edges touches point Q. Draw a line XY\n\n**fig-09.** Line $AB$ with $XY$ now drawn across
```

```
it, $X$ above and $Y$ below, and a right-angle\nmark in the upper right quadrant. The foot of
$XY$ is not labelled $Q$.\n\n[drawing attached to this section as fig-09, marked A, B, X, Y]"
 }
]
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — bisect a given line segment

objective_text

bisect a given line segment

current_step

3

instructional_events

• present_content
• elicit_performance

student_tasks

• construct line segment

source_figures


```
[
 {
  "caption": "The line segment $AB$, a stroke at each end, $A$ on the left and $B$ on the
right.",
  "figure_ref": "fig-10"
 },
 {
  "caption": "The segment $AB$ with arc pairs crossing above and below it, the upper crossing
labelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc from $A$
meets an arc from $B$.",
  "figure_ref": "fig-11"
 },
 {
  "caption": "The same arcs with $P$ joined to $Q$, the joining line crossing $AB$ at its middle
and carrying a right-angle mark in the upper right quadrant.",
  "figure_ref": "fig-12"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedures:**\n\n1. Open a pair of compasses so that the radius is about $\
\frac{3}{4}$ of the length of AB.\n2. To bisect the line segment AB means to divide it into two
equal halves\n\n**fig-10.** The line segment $AB$, a stroke at each end, $A$ on the left and $B$
on the right.\n\n[drawing attached to this section as fig-10, marked A, B]\n\n3. Place the sharp
point of the compass on A, Draw two arcs, one above and the other below the\n   midle of AB.
\n\n**fig-11.** The segment $AB$ with arc pairs crossing above and below it, the upper
crossing\nlabelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc
from $A$ meets\nan arc from $B$.\n\n[drawing attached to this section as fig-11, marked A, B, P,
Q]\n\n4. With the same radius, and place the sharp point of the compasses on B Draw two arc so
that they\n   cut the first arc at P and Q\n\n**fig-12.** The same arcs with $P$ joined to $Q$,
the joining line crossing $AB$ at its middle and\ncarrying a right-angle mark in the upper right
quadrant.\n\n[drawing attached to this section as fig-12, marked A, B, P, Q]"
```

```
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, assessment_points, source_evidence, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 4 — construct angle 90°

`objective_text`

construct angle 90°

`current_step`

4

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct angle 90 degrees

`source_figures`

```
[
 {
  "caption": "The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either side of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle mark at its foot and the two arcs crossing at $Z$ near its top.",
  "figure_ref": "fig-13"
 }
]
```

`lesson_content`

```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedure:**\n\n1. Draw a line segment $AB$ and mark a point $O$ on it where the right angle is to stand.\n2. With centre $O$ and any convenient radius, draw an arc to cut $AB$ at $X$ and $Y$.\n3. With centre $X$ and a radius greater than $XO$, draw an arc above $AB$.\n4. With centre $Y$ and the same radius as in (3), draw an arc to cut the first arc at $Z$.\n5. Join $OZ$. Then $\\angle ZOA = \\angle ZOB = 90^\\circ$.\n\n**fig-13.** The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either\nside of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle\nmark at its foot and the two arcs crossing at $Z$ near its top.\n\n[drawing attached to this section as fig-13, marked A, B, X, Y, O, Z]"
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, assessment_points, source_evidence, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 5 — construct angle 60°

objective_text

construct angle 60°

current_step

5

instructional_events

• present_content
• elicit_performance

student_tasks

• construct angle 60 degrees

source_figures


```
[
 {
  "caption": "The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc swung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a straight line from $B$ to that crossing. The crossing itself is not labelled.",
  "figure_ref": "fig-14"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedure:**\n\n1. Draw a line segment BC with centre B at any convinent radius. Draw an arc to cut BC at X\n2. with centre X and the same radius as in (i) draw an arc to cut the first arc at A\n3. Join AB.\n\n**fig-14.** The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc\nswung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a\nstraight line from $B$ to that crossing. The crossing itself is not labelled.\n\n[drawing attached to this section as fig-14, marked B, C, X]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Constructing a Parallel Line through a Point

EXPLANATION

You already know how to draw straight lines. To draw a line through a given point \(P\) that is parallel to an existing line \(AB\), use a set-square and a ruler:

1. Place the set-square so that one of its edges lies exactly on the line \(AB\) (see fig-01). The right-angle corner of the set-square should be at the left end of that edge.
2. Take a ruler and lay it along one of the other edges of the set-square – the edge that is perpendicular to the one on \(AB\) (fig-02). If you are right-handed, use the left-hand edge of the set-square.
3. Keep the ruler fixed. Slide the set-square along the ruler, keeping it in the same orientation, until the edge that was on \(AB\) reaches the point \(P\) (fig-04 and fig-05). The set-square does not turn; it only translates.
4. Draw a line along that edge of the set-square through \(P\). Because the set-square has not rotated, this new line is parallel to \(AB\).

The important idea is that the set-square provides a fixed angle. By moving it along a straight ruler without rotating, the edge that was on \(AB\) stays parallel to \(AB\) wherever it is placed.

WORKED_EXAMPLE

Worked example

Given: line \(AB\) is drawn horizontally and point \(P\) is above the line as shown in fig-01.

Step 1: Place the set-square so that its bottom edge coincides with \(AB\) (fig-01).

Step 2: Lay a ruler along the vertical edge of the set-square (the edge that is perpendicular to \(AB\)) – this is the edge shown in fig-02.

Step 3: Hold the ruler firmly. Slide the set-square straight up the ruler until the bottom edge of the set-square passes through \(P\). The set-square moves without turning (fig-04 and fig-05).

Step 4: Using a pencil, draw a line along the bottom edge of the set-square now passing through \(P\). The line you have drawn is parallel to \(AB\).

Check: The two lines never meet, and the distance between them is the same at every point – this confirms they are parallel.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \node[left] at (0,0) {$A$};
  \node[above] at (7,0) {$B$};
  \fill (3.4,2.8) circle (1.6pt);
  \node[above] at (3.4,2.8) {$P$};
  \draw (2.7,0) -- (2.7,-1.1) -- (5.2,0);
  \draw (2.7,-0.26) -- (2.96,-0.26) -- (2.96,0);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw[rounded corners=3pt] (1.29,3.50) rectangle (1.72,-4.12);
  % The set square's vertical edge lies ALONG the ruler's right edge (x=1.72) — in the
  % source the two are layered, so only one line is visible. Drawn collinear, not offset.
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above right] at (8,0) {$B$};
  % The ruler now lies along the sloping edge, extending past it at both ends. Drawn a
  % hairline clear of that edge so the set square stays visible as a triangle.
  \draw[rounded corners=2pt] (-0.171,-3.033) -- (6.760,0.545) -- (6.967,0.145) -- (0.036,-3.433) -- cycle;
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw (0.71,2.40) -- (8,2.40);
  \draw[rounded corners=3pt] (0.28,3.65) rectangle (0.71,-3.97);
  % ONE set square drawn in two positions, translated straight up the ruler's right edge
  % (x=0.71). Both are congruent — leg 1.21, width 2.68 — so the two hypotenuses are
  % exactly parallel, which is the visible signature of a slide.
  \draw (0.71,0) -- (0.71,-1.21) -- (3.39,0);
  \draw (0.71,-0.26) -- (0.97,-0.26) -- (0.97,0);
  \draw (0.71,2.40) -- (0.71,1.19) -- (3.39,2.40);
  \draw (0.71,2.14) -- (0.97,2.14) -- (0.97,2.40);
  \draw[->] (2.44,0.1) -- (2.44,1.37);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above right] at (8,0) {$B$};
  \draw (0.35,2.2) -- (8,2.2);
  \fill (6.0,2.2) circle (1.6pt);
  \node[above] at (6.0,2.2) {$P$};
  % The ruler is held still; it spans both positions of the set square, drawn a hairline
  % clear of their sloping edges.
  \draw[rounded corners=2pt] (-0.026,-2.119) -- (7.616,2.247) -- (7.824,1.882) -- (0.182,-2.484) -- cycle;
  \draw (0.6,0) -- (0.6,-1.6) -- (3.4,0);
  \draw (0.6,-0.26) -- (0.86,-0.26) -- (0.86,0);
  % The same set square, translated along the ruler until its edge reaches P.
  \draw (4.45,2.2) -- (4.45,0.6) -- (7.25,2.2);
  \draw (4.45,1.94) -- (4.71,1.94) -- (4.71,2.2);
  \draw[->] (2.6,0.55) -- (4.0,1.35);
\end{tikzpicture}
```

1. Line \(CD\) is drawn on your paper. Point \(Q\) lies somewhere above the line. Using a set-square and a ruler, construct a line through \(Q\) that is parallel to \(CD\). Describe the steps you would take.

answer: 1. Place the set-square with one edge on \(CD\). 2. Lay a ruler along a perpendicular edge of the set-square. 3. Keep the ruler fixed and slide the set-square up the ruler until the edge on \(CD\) reaches \(Q\). 4. Draw a line along that edge through \(Q\).

2. You have drawn line \(EF\) and a point \(R\) to the right of the line. Construct a line through \(R\) parallel to \(EF\). List the actions you perform with the set-square and ruler.

answer: 1. Align a set-square edge with \(EF\). 2. Place a ruler along the set-square's vertical edge. 3. Slide the set-square along the ruler until the aligned edge reaches \(R\). 4. Draw the new line through \(R\) using that edge.

How to Construct a Perpendicular Line

EXPLANATION

A line is perpendicular to another when the two lines meet at a right angle (\(90^\circ\)). To draw a line that is perpendicular to a given line \(AB\) at a point \(Q\) on \(AB\), you can use a set square:

1. Draw the base line \(AB\) and mark the point \(Q\) where you want the perpendicular line to start (see fig-07).
2. Place the edge of the set square on the line \(AB\) so that one side of the set square lies along \(AB\). The right-angle corner of the set square should be at the point where the set square meets the line (as shown in fig-08).
3. Keeping the set square pressed firmly, slide it along the hypotenuse side until the opposite edge of the set square touches the point \(Q\).
4. Draw a line through \(Q\) using the upright edge of the set square. This new line \(XY\) will be perpendicular to \(AB\) (see fig-09).

The right-angle mark in the diagram confirms that the two lines meet at \(90^\circ\).

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.15,-0.1) {$A$};
  \node[above] at (4.52,0.1) {$B$};
  \draw (2.1,1.85) -- (2.15,-0.8);
  \node[above] at (2.1,1.85) {$X$};
  \node[below] at (2.15,-0.8) {$Y$};
  \draw (2.44,0) -- (2.44,0.32) -- (2.12,0.32);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.41,-0.05) {$A$};
  \node[above] at (3.93,0.05) {$B$};
  % The upright edge stands square on the base: step 2 and 3 name a hypotenuse and a
  % "third edge", which only a marked right angle lets the reader tell apart.
  \draw (1.54,0) -- (2.25,1.0) -- (2.25,0) -- cycle;
  \draw (1.97,0) -- (1.97,0.28) -- (2.25,0.28);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below] at (0,-0.28) {$A$};
  \node[below] at (7,-0.28) {$B$};
\end{tikzpicture}
```

WORKED_EXAMPLE

Worked example

You have a line \(AB\) that is 8 cm long. You need to construct a line perpendicular to \(AB\) at a point \(Q\) that is 3 cm from \(A\).

1. Draw \(AB\) and label the ends \(A\) and \(B\). Mark point \(Q\) on \(AB\) such that \(AQ = 3\) cm.
2. Place the set square on \(AB\) with its base along \(AB\) and the right-angle corner at the point where the set square meets the line.
3. Slide the set square down the hypotenuse until the opposite edge touches \(Q\).
4. Using the upright edge of the set square, draw the line \(XY\) through \(Q\).

The line \(XY\) is now perpendicular to \(AB\) at \(Q\).

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-0.5,0) -- (7.5,0);
```

```
  \draw (0,0.25) -- (0,-0.25);
  \draw (5,0.25) -- (5,-0.25);
  \node[below] at (0,-0.3) {$A$};
  \node[below] at (5,-0.3) {$B$};
  \draw (2.5,2.6) -- (2.5,-2.6);
  \node[above] at (2.5,2.6) {$X$};
  \node[below] at (2.5,-2.6) {$Y$};
  \draw (2.82,0) -- (2.82,0.32) -- (2.5,0.32);
\end{tikzpicture}
```

1. Draw a line \(CD\) that is 10 cm long. Mark a point \(P\) 4 cm from \(C\). Construct a line through \(P\) that is perpendicular to \(CD\).

answer: A line drawn through \(P\) using the upright edge of the set square will be perpendicular to \(CD\).

2. A line \(EF\) is drawn horizontally. You need a perpendicular line that passes through a point \(R\) located exactly in the middle of \(EF\). Describe the steps you would take.

answer: 1. Mark the midpoint \(R\) on \(EF\). 2. Place the set square on \(EF\) with its base along \(EF\). 3. Slide the set square until the opposite edge touches \(R\). 4. Draw the line through \(R\) using the upright edge; this line is perpendicular to \(EF\).

3. Given a line \(GH\) and a point \(S\) not on the line, can you use the set square method described to construct a perpendicular line from \(S\) to \(GH\)? Explain why or why not.

answer: No. The set square method requires the point to be on the given line. Since \(S\) is not on \(GH\), you must first draw a line from \(S\) to \(GH\) (e.g., using a ruler) and then apply the set square method at the intersection point.

How to Bisect a Line Segment

> EXPLANATION
> To find the exact middle of a line segment AB, follow these steps:
>
> 1. Open your compasses to a width a little less than three-quarters of the length of AB.
> 2. Place the compass point on endpoint **A** and draw an arc above the line and another arc below it.
> 3. Without changing the compass width, place the point on endpoint **B** and draw two arcs that intersect the first pair. The upper intersection is labelled **P** and the lower one **Q** (see fig-11).
> 4. Use a ruler to join **P** and **Q**. The line **PQ** will cross AB at its midpoint **M** (fig-12).
> 5. The point **M** is the bisector: the two new segments **AM** and **MB** are equal in length, and **PQ** meets AB at a right angle.
>
> Remember: the same compass width is used for both ends, and the arcs must intersect to give clear points **P** and **Q**.

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (0,0) -- (7,0);
>  \draw (0,0.22) -- (0,-0.22);
>  \draw (7,0.22) -- (7,-0.22);
>  \node[below left] at (0,-0.1) {$A$};
>  \node[below right] at (7,-0.1) {$B$};
> \end{tikzpicture}
> ```

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (0,0) -- (6,0);
>  \draw (0,0.22) -- (0,-0.22);
>  \draw (6,0.22) -- (6,-0.22);
>  \node[below left] at (0,-0.1) {$A$};
>  \node[below right] at (6,-0.1) {$B$};
>  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
>  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
>  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
>  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
>  \node[above] at (3,3.5) {$P$};
>  \node[below] at (3,-3.5) {$Q$};
> \end{tikzpicture}
> ```

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (0,0) -- (6,0);
>  \draw (0,0.22) -- (0,-0.22);
>  \draw (6,0.22) -- (6,-0.22);
>  \node[below left] at (0,-0.1) {$A$};
>  \node[below right] at (6,-0.1) {$B$};
>  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
>  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
>  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
>  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
>  \draw (3,3.35) -- (3,-3.35);
>  \node[above] at (3,3.35) {$P$};
>  \node[below] at (3,-3.35) {$Q$};
>  \node[below right] at (3,-0.1) {$M$};
>  \draw (3.32,0) -- (3.32,0.32) -- (3,0.32);
> \end{tikzpicture}
> ```

1. You have a line segment CD that is 8 cm long. Using the method described, where will the midpoint E be located on the segment? State the lengths of CE and ED.

answer: CE = 4 cm and ED = 4 cm

2. A line segment FG is drawn. After constructing the arcs, the points of intersection are H (above) and I (below). The line HI meets FG at point J. If FG is 12 cm long, what is the length of FJ?

answer: FJ = 6 cm

Objective 4

Constructing a Right Angle (90°)

EXPLANATION
To draw a right angle at a point on a line you need only a ruler and a compass.

1. Draw a straight line \(AB\) and pick the point \(C\) on it where the \(90^\circ\) angle is required.
2. Place the compass point at \(C\) and draw an arc that cuts the line at two points; label them \(D\) and \(E\).
3. Without changing the compass width, move the point to \(D\) and draw a larger arc above the line.
4. Keeping the same width, place the compass at \(E\) and draw another arc that meets the previous one at \(F\).
5. Join \(C\) to \(F\) with a straight line.
6. The line \(CF\) is perpendicular to \(AB\); therefore \(\angle FCB = 90^\circ\).

Refer to figure fig-13 for a visual guide of the construction.

VISUAL
```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
  \draw (0.29,0.22) -- (0.29,-0.22);
  \draw (0.84,0.34) -- (0.84,-0.34);
  \draw (6.13,0.22) -- (6.13,-0.22);
  \draw (6.60,0.34) -- (6.60,-0.34);
  \draw (3.8,0) -- (3.8,3.2);
  \draw (4.12,0) -- (4.12,0.32) -- (3.8,0.32);
  \draw (2.9,2.55) -- (4.7,3.05);
  \draw (2.9,3.05) -- (4.7,2.55);
\end{tikzpicture}
```

WORKED_EXAMPLE
Worked Example

Suppose you have a line \(AB\) 8 cm long and you need a right angle at a point \(C\) 3 cm from \(A\).

- Draw \(AB\) and mark \(C\) as described.
- With the compass set to any convenient radius, draw an arc from \(C\) intersecting \(AB\) at \(D\) and \(E\).
- Increase the compass opening a little beyond \(CD\). From \(D\) draw an arc above the line.
- With the same opening, draw an arc from \(E\) that meets the first arc at \(F\).
- Connect \(C\) to \(F\). The line \(CF\) forms a perfect right angle with \(AB\).

This demonstrates that the exact size of the arcs is not important; only that the second set of arcs intersect.

1. You have a line \(XY\) and need to construct a right angle at point \(Z\) on the line. List the steps you would follow.
answer: 1. Draw line XY and mark point Z. 2. With compass at Z draw an arc cutting XY at points P and Q. 3. Widen the compass, draw an arc from P above the line. 4. With same width draw an arc from Q intersecting the previous arc at R. 5. Join Z to R; ZR is perpendicular to XY, giving a 90° angle.

2. A line \(MN\) is drawn and a point \(O\) is chosen on it. After constructing the perpendicular at \(O\), you obtain point \(P\) where the new line meets the first arc. What is the measure of \(\angle MON\)?
answer: 90°

Objective 5

Constructing a 60° Angle

> EXPLANATION
> To draw a 60° angle you only need a straightedge and a compass.
>
> 1. Draw a line segment BC of any convenient length. Mark a point X on BC (see fig-14).
> 2. Place the compass point at B and draw an arc that cuts BC at X.
> 3. Without changing the compass width, place the compass point at X and draw a second arc that cuts the first arc at a point A.
> 4. Join A to B with a straight line. The angle $\angle ABX$ is exactly 60°.
>
> The construction works because the two arcs create an equilateral triangle ABX, and each interior angle of an equilateral triangle is 60°.

> VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \node[below left] at (0,-0.05) {$B$};
  \node[above right] at (7,0) {$C$};
  \draw (5.17,0.22) -- (5.17,-0.22);
  \node[below] at (5.17,-0.25) {$X$};
  \draw (0,0) ++(0:5.17) arc (0:74:5.17);
  \draw (5.17,0) ++(103:5.17) arc (103:140:5.17);
  \draw (0,0) -- (2.59,4.48);
\end{tikzpicture}
```

> WORKED_EXAMPLE
> **Worked example**
>
> - Draw BC 8 cm long and label the ends B (left) and C (right).
> - Mark X 5 cm from B on BC.
> - Set the compass to a radius of 5 cm (the distance BX) and draw an arc from B through X.
> - Keeping the same radius, place the compass at X and draw an arc that meets the first arc at A.
> - Connect A to B.
>
> The angle $\angle ABX$ you have just drawn is a perfect 60° angle.

1. Using a ruler and compass, construct a 60° angle on a fresh sheet of paper. Describe each step you take.

answer: Step-1: Draw a line segment BC and mark a point X on it. Step-2: With centre B draw an arc cutting BC at X. Step-3: With the same radius, centre the compass at X and draw an arc intersecting the first arc at A. Step-4: Join A to B. The angle ABX is 60°.

2. Given a line segment DE of length 10 cm, construct a 60° angle at D. What is the length of the side of the equilateral triangle formed in the construction?

answer: The side length is 10 cm, because the arcs are drawn with radius DE, forming an equilateral triangle D-E-F where DF = DE = 10 cm.

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	5	Scope	**grounded**
Explanation the principle, rule or relationship to review	5	Enacted content	**grounded**
Worked example a problem with its solution worked through	4	Enacted content	**grounded**

Figure a drawing the learner reads from	14 Figures	**grounded**
Practice what the learner does	11 Set work	**grounded**
Answer what the learner checks against	11 Enacted content	**grounded**